Problem 1

$$F(x,y,z) = x^2 + xy + y^2 + z^2, \quad P = (1,1,1), \quad Q = (3,0,3).$$

1. Use a unit direction vector to calculate the directional derivative at P toward Q, stating the dot-product formula you use.
2. Use the gradient to give the unit direction of fastest increase at P.
3. Use the gradient to find the tangent plane to the level surface $F(x,y,z) = F(P)$ at P, first evaluating $F(P)$.

Problem 2

$$f(x, y) = x^3 - 3x + y^2, \qquad R = [-2, 2] \times [-1, 1].$$

1. Write and solve the equations for all critical points of f.
2. State the Second Derivatives Test and use it to classify the critical points.
3. Find the absolute maximum and minimum on R by checking its boundary as well as its interior.

Problem 3

$$f(x, y) = xy, \qquad x^2 + 4y^2 = 8.$$

1. Set up the complete Lagrange multiplier system, including the constraint.
2. Find the maximum and minimum by comparing the values at all candidate points.

Problem 4

$$I = \int_0^1 \int_{\sqrt[3]{x}}^1 e^{y^4}\, dy\, dx.$$

1. Set up the reversed integral using horizontal slices of the region.
2. Evaluate it, explaining the factor produced by integrating with respect to x.

Problem 5

Let $D = \{(x, y) : y \geq 0,\ x^2 + y^2 \leq 4x\}$. Let V be the volume under $z = y$ and above D.

1. Set up the polar-coordinate volume integral, showing both the height and the area element.
2. Evaluate the volume.

Problem 6

Let E be the first-octant solid bounded by $x + y + z = 2$.

$$I = \iiint_E z \, dV.$$

1. Set up I in the order $dy\,dz\,dx$ by first fixing (x, z) and finding the y-segment.
2. Evaluate the resulting integral.

Problem 7

Let E lie above $z = x^2 + y^2$ and below $z = 2\sqrt{x^2 + y^2}$.

1. Set up the cylindrical-coordinate volume integral from the two surfaces and their intersection.
2. Evaluate the volume.

Problem 8

Let E lie inside $x^2 + y^2 + z^2 = 4z$ and above $z = \sqrt{x^2 + y^2}$.

1. Set up the spherical-coordinate volume integral by converting both the sphere and the cone.
2. Evaluate the volume.

Problem 9

Let R be the first-quadrant region bounded by

$$xy = 1, \quad xy = 2, \quad y = x, \quad y = 2x.$$

$$I = \iint_R \frac{y}{x} e^{xy}\, dA.$$

1. Choose new variables from the boundary curves that transform R into a rectangle.
2. Derive the inverse transformation using the first-quadrant condition.
3. Find the area element dA from the Jacobian of the inverse transformation.
4. Transform the whole integral, showing all factors before simplifying, and evaluate it.